IDW and ILDW (Short Pseudocode)

Notation. Each link ℓ is the segment $[p_{\ell,1}, p_{\ell,2}]$ with endpoints $p_{\ell,1}, p_{\ell,2} \in \mathbb{R}^2$, midpoint $m_\ell = (p_{\ell,1} + p_{\ell,2})/2$, and length $L_\ell = \|p_{\ell,2} - p_{\ell,1}\|_2$. Each pixel p has center $c_p \in \mathbb{R}^2$.

IDW

Algorithm 1 Discretized Midpoint IDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_\ell, f_\ell, \mathrm{pol}_\ell)$, radius $r_{\max}$

for each link ℓ **do**
    $L_\ell \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$ ▷ link length
    $(\kappa_\ell, \alpha_\ell) \leftarrow \mathrm{ITU838}(f_\ell, \mathrm{pol}_\ell)$
    $R_\ell \leftarrow \left(\dfrac{A_\ell / L_\ell}{\kappa_\ell}\right)^{1/\alpha_\ell}$ ▷ link rain estimate
end for
for each pixel p with center c_p **do**
    $m_\ell \leftarrow (p_{\ell,1} + p_{\ell,2})/2$ for each link ℓ ▷ link midpoint
    if one or more link midpoints lie inside pixel p **then**
        $\hat{R}_p \leftarrow$ mean of R_ℓ of those links ▷ inside-pixel average
    else
        $N_p \leftarrow \{\ell : \|m_\ell - c_p\|_2 \le r_{\max}\}$ ▷ nearby links
        if $N_p = \emptyset$ **then**
            $\hat{R}_p \leftarrow 0$
        else
            $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \dfrac{\|m_\ell - c_p\|_2^{-2}}{\sum_{j \in N_p} \|m_j - c_p\|_2^{-2}} R_\ell$ ▷ inverse-distance weighting
        end if
    end if
end for
Return field $\hat{R}$

ILDW

Algorithm 2 Segment-Distance ILDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_\ell, f_\ell, \mathrm{pol}_\ell)$, radius $r_{\max}$

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for each link ℓ do
    L_ℓ ← ‖p_{ℓ,2} − p_{ℓ,1}‖_2                              ▷ link length
    (κ_ℓ, α_ℓ) ← ITU838(f_ℓ, pol_ℓ)
    R_ℓ ← ((A_ℓ/L_ℓ) / κ_ℓ)^{1/α_ℓ}                          ▷ link rain estimate
end for
for each pixel p with center c_p do
    if one or more links cross pixel p then
        R̂_p ← mean of R_ℓ of those links                     ▷ crossing-link average
    else
        N_p ← {ℓ : dist(c_p, [p_{ℓ,1}, p_{ℓ,2}]) ≤ r_max}     ▷ nearby links
        if N_p = ∅ then
            R̂_p ← 0
        else
            R̂_p ← Σ_{ℓ∈N_p} dist(c_p,[p_{ℓ,1},p_{ℓ,2}])^{−2} / Σ_{j∈N_p} dist(c_p,[p_{j,1},p_{j,2}])^{−2} · R_ℓ   ▷ inverse-distance weighting
        end if
    end if
end for
Return field R̂
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Where:

$$L_\ell \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$$

$$R_\ell \leftarrow \left(\frac{A_\ell / L_\ell}{\kappa_\ell}\right)^{1/\alpha_\ell}$$

$$N_p \leftarrow \{\ell : \mathrm{dist}(c_p, [p_{\ell,1}, p_{\ell,2}]) \le r_{\max}\}$$

$$\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\mathrm{dist}(c_p, [p_{\ell,1}, p_{\ell,2}])^{-2}}{\sum_{j \in N_p} \mathrm{dist}(c_p, [p_{j,1}, p_{j,2}])^{-2}} R_\ell$$